

Monte Carlo Methods for Basket Option Pricing

with Variance Reduction Techniques

Louen Marx

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Abstract

This project implements and compares several Monte Carlo simulation methods for pricing European basket call options written on two correlated assets following geometric Brownian motion. We study a standard Monte Carlo estimator based on the plain discounted payoff, a reduced-variance estimator that exploits conditional expectation to analytically integrate out one source of randomness, and two control variate approaches — one relying on put-call parity and the other using the geometric basket as a correlated control variate with a known closed-form price.

Each method is analyzed in terms of pricing accuracy, standard deviation, and convergence behavior across a range of parameters including asset correlation ρ , strike price K , and basket weights α, β . We also compute the delta sensitivity of the option via centered finite differences and compare the result against the analytical log-normal approximation.

The results demonstrate that variance reduction techniques can dramatically improve computational efficiency. The reduced-variance estimator based on conditional expectation achieves variance reductions of several orders of magnitude compared to the standard Monte Carlo approach, while the geometric basket control variate proves especially powerful for out-of-the-money options where the standard estimator struggles most. The put-call parity control variate, on the other hand, is most effective for in-the-money options. Taken together, these methods form a complementary toolkit: no single technique dominates across all parameter regimes, but a well-chosen method can reduce the standard error by a factor of 10 or more relative to the crude Monte Carlo estimator.

Keywords: Monte Carlo simulation, basket options, variance reduction, conditional expectation, control variates, Black-Scholes approximation, delta hedging, Cholesky decomposition

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1 Introduction

1.1 Context and Motivation

Basket options are among the most widely traded exotic derivatives in financial markets. They are commonly used in equity, commodity, and foreign exchange markets to hedge exposure to a weighted portfolio of assets. Unlike plain vanilla options on a single underlying, basket options derive their payoff from a linear combination of several assets, making them both more flexible as hedging instruments and more challenging to price accurately.

The fundamental difficulty in pricing basket options lies in a well-known mathematical fact: the sum of log-normal random variables is **not** itself log-normally distributed. This means that the celebrated Black-Scholes formula, which assumes a single log-normal underlying, cannot be applied directly. There is no general closed-form solution for the price of a basket option, and practitioners must therefore turn to either analytical approximations or numerical methods.

Among numerical approaches, **Monte Carlo simulation** stands out as the most flexible and general-purpose technique. The core idea is disarmingly simple: simulate a large number of possible future scenarios for the asset prices, compute the option payoff in each scenario, and average the results. Under the risk-neutral measure, this sample average converges to the true option price as the number of simulations grows, by virtue of the law of large numbers. The method is remarkably flexible — it handles arbitrary payoff structures, multiple assets, stochastic volatility, and path-dependent features with essentially no modification to the core algorithm.

However, this flexibility comes at a cost. The main drawback of Monte Carlo methods is their **slow rate of convergence**: the standard error of the estimator decreases only as $O(1/\sqrt{N})$, where N is the number of simulated paths. In concrete terms, to halve the estimation error, one must quadruple the number of simulations. For high-precision pricing — for instance, when computing deltas for hedging, or when marking-to-market a large book of derivatives to within cent accuracy — this can require millions of simulated paths, which is computationally expensive.

This fundamental trade-off between flexibility and efficiency motivates the development and study of **variance reduction techniques**. These are methods that preserve the unbiasedness of the Monte Carlo estimator while significantly reducing its variance. The key insight is that not all simulations are equally informative: some carry more information about the option price than others. By cleverly restructuring the estimator — for example, by analytically integrating out part of the randomness, or by exploiting known relationships between correlated quantities — one can extract more information from each simulated path, thereby achieving the same precision with far fewer simulations.

1.2 Problem Statement

In this project, we consider a European call option on a basket of two correlated assets S_1 and S_2 . Under the risk-neutral measure $\mathbb{Q}$, each asset follows geometric Brownian motion:

$$S_1(T) = S_1(0) \exp \left[\left(r - \frac{\sigma_1^2}{2} \right) T + \sigma_1 W_1(T) \right] \quad (1)$$

$$S_2(T) = S_2(0) \exp \left[\left(r - \frac{\sigma_2^2}{2} \right) T + \sigma_2 W_2(T) \right] \quad (2)$$

where r is the risk-free interest rate, σ_1 and σ_2 are the respective volatilities, T is the time to maturity, and $W_1(T)$, $W_2(T)$ are Brownian motions at time T under $\mathbb{Q}$, with correlation $\text{Corr}(W_1, W_2) = \rho$.

The parameter ρ plays a central role in this model. It captures the degree of co-movement between the two assets: when $\rho > 0$, the assets tend to move in the same direction; when $\rho < 0$, they tend to move in opposite directions. From a financial perspective, ρ determines the

diversification benefit of the basket — negatively correlated assets partially offset each other's fluctuations, reducing the overall volatility of the basket and thus the option price.

To simulate the correlated Brownian motions, we use the **Cholesky decomposition** of the correlation matrix. The 2×2 correlation matrix is:

$$\Gamma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = A A^\top \quad (3)$$

where the lower-triangular Cholesky factor is:

$$A = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{pmatrix} \quad (4)$$

The correlated Brownian increments are then obtained as $W = \sqrt{T} A G$, where $G = (X, Y)$ is a vector of independent standard normal variables. This guarantees that $\text{Cov}(W_1, W_2) = \rho T$, as required.

The independent standard normals X and Y are generated via the classical **Box-Muller transform**:

$$X = \sqrt{-2 \ln U} \cos(2\pi V), \quad Y = \sqrt{-2 \ln U} \sin(2\pi V) \quad (5)$$

with $U, V \sim \text{Uniform}(0, 1)$ drawn independently. This method produces pairs of independent standard normal variables from pairs of uniform random numbers, and has the advantage of being exact (it does not rely on the central limit theorem or any approximation).

The payoff of the European basket call option at maturity is:

$$\text{Payoff} = e^{-rT} \max(\alpha S_1(T) + \beta S_2(T) - K, 0) \quad (6)$$

where α and β are the basket weights (determining the contribution of each asset to the basket) and K is the strike price. The fair price of the option, under the risk-neutral pricing framework, is simply the discounted expectation of this payoff:

$$P = \mathbb{E}^\mathbb{Q}[e^{-rT} \max(\alpha S_1(T) + \beta S_2(T) - K, 0)] \quad (7)$$

This expectation cannot be computed in closed form because the argument of the max function involves a sum of log-normal random variables, whose distribution is analytically intractable.

1.3 Objectives

The goals of this project are as follows:

1. Implement a **log-normal approximation** based on moment matching to obtain a fast analytical price estimate, and assess its accuracy.
2. Implement a **standard Monte Carlo** estimator and analyze its convergence properties, including the behavior of confidence intervals as the sample size grows.
3. Develop a **reduced-variance estimator** using conditional expectation, which analytically integrates out one source of randomness by conditioning on one of the Brownian motions.
4. Implement two **control variate** methods: one based on put-call parity (pricing the call via the put payoff), and the other using the geometric basket as a highly correlated control with a known closed-form price.
5. Study the **sensitivity** of the option price and the effectiveness of each variance reduction technique with respect to key parameters: correlation ρ , strike price K , and basket weight α .

6. Compute the **delta** sensitivity ($\Delta_1 = \partial P / \partial S_1(0)$) via centered finite differences and compare the result across methods.

1.4 Default Parameters

Unless stated otherwise, the following parameters are used throughout this study:

Parameter	Symbol	Value
Initial prices	$S_1(0), S_2(0)$	1, 1
Volatilities	σ_1, σ_2	0.35, 0.40
Risk-free rate	r	0.01
Correlation	ρ	0.3
Maturity	T	2 years
Strike	K	2
Basket weights	α, β	1, 1
Sample size	N	1 000 000

Table 1: Default model parameters used throughout the study.

With these parameters, the basket initial value is $S_B(0) = \alpha S_1(0) + \beta S_2(0) = 2$ and the strike is $K = 2$, making the option approximately at-the-money at inception. The two-year maturity and relatively high volatilities ($\sigma_1 = 35\%$, $\sigma_2 = 40\%$) result in a significant time value. The correlation $\rho = 0.3$ reflects a moderate positive co-movement between the two assets, which is realistic for assets within the same market sector.

The choice of equal initial prices ($S_1(0) = S_2(0) = 1$) and equal weights ($\alpha = \beta = 1$) simplifies the analysis by treating both assets symmetrically. However, we will vary these parameters in Sections 6–8 to study the sensitivity of our results.

2 Normal CDF Approximation

Many of the analytical formulas used in this project — the Black-Scholes price, the conditional expectation formula, the geometric basket price — require evaluating the standard normal cumulative distribution function $\Phi(x)$. While the SciPy library provides a highly accurate implementation of $\Phi(x)$ based on rational approximations with machine-precision accuracy, we also implement a standalone polynomial approximation for verification purposes and to understand the numerical foundations of the method.

2.1 Abramowitz & Stegun Approximation

The approximation we use is based on the classical formula from Abramowitz & Stegun’s *Handbook of Mathematical Functions* (1964). It is a rational (polynomial) approximation that achieves a maximum absolute error of 7.5×10^{-8} , which is more than sufficient for financial applications where prices are typically quoted to 4–6 decimal places.

For $x \geq 0$, the approximation is:

$$\Phi(x) = 1 - \varphi(x) (b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5), \quad t = \frac{1}{1 + b_0 x} \quad (8)$$

where $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the standard normal probability density function (PDF) and the coefficients are:

$$\begin{aligned}
b_0 &= 0.2316419, & b_1 &= 0.319381530, & b_2 &= -0.356563782 \\
b_3 &= 1.781477937, & b_4 &= -1.821255978, & b_5 &= 1.330274429
\end{aligned}$$

2.2 Properties and Implementation

This approximation exploits several key properties of the normal distribution:

- **Symmetry:** For $x < 0$, we use $\Phi(x) = 1 - \Phi(-x)$, which reduces the problem to evaluating Φ only for non-negative arguments.
- **Decay:** The factor $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ ensures rapid decay for large $|x|$, so the polynomial correction only needs to be accurate in a bounded region.
- **Efficiency:** The change of variable $t = 1/(1 + b_0x)$ maps $[0, \infty)$ to $(0, 1]$, making the polynomial well-conditioned across the entire range.

The polynomial in t provides an efficient and accurate way to evaluate the CDF without numerical integration, making it well-suited for repeated evaluation inside Monte Carlo loops. In our implementation, we use this custom approximation as a cross-check against the SciPy function `scipy.stats.norm.cdf`, which is used in the main pricing functions for maximum accuracy.

3 Log-Normal Approximation (Black-Scholes)

3.1 The Log-Normality Problem

Since the arithmetic basket $B(T) = \alpha S_1(T) + \beta S_2(T)$ is a weighted sum of log-normal random variables, it is **not** itself log-normally distributed. This is a fundamental obstacle: the Black-Scholes formula requires the underlying to be log-normal, and there is no way to make the arithmetic basket fit this assumption exactly. The distribution of $B(T)$ is in fact positively skewed with heavier tails than a log-normal distribution, because the sum of exponentials is dominated by the largest terms.

3.2 Moment Matching Approach

However, a common and useful approximation consists in **fitting a log-normal distribution to the basket** by matching its first two moments. The idea is to find an “equivalent” single asset S_B that is log-normally distributed and has the same first and second moments as the basket $B(T)$. We then apply the standard Black-Scholes formula to this equivalent asset.

We define the equivalent basket with initial price and risk-free rate:

$$S_B(0) = \alpha S_1(0) + \beta S_2(0), \quad r_B = r \quad (9)$$

The first moment is automatically matched since $\mathbb{E}[B(T)] = S_B(0)e^{rT}$ by the martingale property of discounted asset prices under $\mathbb{Q}$. The equivalent basket volatility σ_B is found by matching the second moment. Under the risk-neutral measure, the second moment of the basket is:

$$\mathbb{E}[B(T)^2] = \alpha^2 S_1(0)^2 e^{(\sigma_1^2 + 2r)T} + \beta^2 S_2(0)^2 e^{(\sigma_2^2 + 2r)T} + 2\alpha\beta S_1(0)S_2(0)e^{(\rho\sigma_1\sigma_2 + 2r)T} \quad (10)$$

The three terms correspond, respectively, to the variance contributions of S_1 alone, S_2 alone, and the cross-term due to correlation. Setting this equal to the second moment of a log-normal variable $S_B(0) e^{(r_B - \sigma_B^2/2)T + \sigma_B W(T)}$, which is $S_B(0)^2 e^{(2r_B + \sigma_B^2)T}$, and solving for σ_B yields:

$$\sigma_B = \sqrt{\frac{1}{T} \ln \left(\frac{\alpha^2 S_1(0)^2 e^{\sigma_1^2 T} + \beta^2 S_2(0)^2 e^{\sigma_2^2 T} + 2\alpha\beta S_1(0)S_2(0)e^{\rho\sigma_1\sigma_2 T}}{S_B(0)^2} \right)} \quad (11)$$

This expression has a natural interpretation: the equivalent basket volatility is a weighted geometric average of the individual volatilities and their correlation, capturing the combined risk of the basket.

3.3 Black-Scholes Price

Once the equivalent parameters are determined, the Black-Scholes price is:

$$P_{BS} = S_B(0) \Phi(d_1) - K e^{-r_B T} \Phi(d_2) \quad (12)$$

with

$$d_1 = \frac{\ln(S_B(0)/K) + (r_B + \sigma_B^2/2)T}{\sigma_B \sqrt{T}}, \quad d_2 = d_1 - \sigma_B \sqrt{T} \quad (13)$$

3.4 Limitations

This approximation provides a fast analytical benchmark but is not exact. The log-normal assumption for the basket is violated because the true distribution of $B(T)$ has a heavier right tail and greater positive skewness than a log-normal distribution with the same first two moments. This is a consequence of the convexity of the exponential function: the sum of log-normals tends to be dominated by the largest realizations, creating a fat right tail.

As a result, the approximation tends to underestimate the true price for certain parameter configurations and overestimate it for others. The discrepancy is most pronounced in the following regimes:

- **Extreme correlations** (ρ near ± 1): the true basket distribution departs significantly from log-normality.
- **Extreme strikes** (deep in-the-money or deep out-of-the-money): the option payoff is sensitive to the tails of the distribution, where the log-normal approximation is least accurate.
- **Disparate volatilities**: when σ_1 and σ_2 are very different, the basket distribution becomes more skewed.

We will quantify these errors empirically in Sections 6–8.

4 Standard Monte Carlo Estimator

4.1 Principle

The standard Monte Carlo estimator is the most straightforward approach to pricing the basket option. It relies directly on the law of large numbers: given N independent draws of the asset prices at maturity, the sample mean of the discounted payoffs converges almost surely to the true option price.

For each simulation $i = 1, \dots, N$, the algorithm proceeds in three steps:

1. Generate a pair of independent standard normals $(X^{(i)}, Y^{(i)})$ via the Box-Muller transform.

2. Apply the Cholesky decomposition to obtain correlated Brownian increments, then compute the terminal asset prices $S_1^{(i)}(T)$ and $S_2^{(i)}(T)$ using equations (1)–(2).
3. Evaluate the discounted payoff $e^{-rT} \max(\alpha S_1^{(i)}(T) + \beta S_2^{(i)}(T) - K, 0)$.

The Monte Carlo price estimate is the sample average:

$$\hat{P}_{\text{MC}} = \frac{1}{N} \sum_{i=1}^N e^{-rT} \max(\alpha S_1^{(i)}(T) + \beta S_2^{(i)}(T) - K, 0) \quad (14)$$

By the strong law of large numbers, $\hat{P}_{\text{MC}} \rightarrow P$ almost surely as $N \rightarrow \infty$. Furthermore, by the central limit theorem, the estimator is asymptotically normal:

$$\sqrt{N} (\hat{P}_{\text{MC}} - P) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \quad (15)$$

where $\sigma^2 = \text{Var}(e^{-rT} \max(\alpha S_1(T) + \beta S_2(T) - K, 0))$ is the true variance of the discounted payoff. This result justifies the construction of confidence intervals for the price estimate.

4.2 Confidence Interval Construction

The standard deviation of the simulated payoffs provides a measure of the estimator's precision:

$$\widehat{\text{STD}} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\text{payoff}^{(i)} - \hat{P}_{\text{MC}})^2} \quad (16)$$

The 90% confidence interval half-width is:

$$\text{error}_{90\%} = \frac{1.645 \times \widehat{\text{STD}}}{\sqrt{N}} \quad (17)$$

where 1.645 is the 95th percentile of the standard normal distribution (the two-sided 90% interval uses the 95th percentile on each side). This means that, with approximately 90% confidence (for large N), the true price lies within $\hat{P}_{\text{MC}} \pm \text{error}_{90\%}$.

The width of this confidence interval is directly proportional to the standard deviation of the payoff and inversely proportional to $\sqrt{N}$. This is the fundamental relationship that motivates variance reduction: if we can reduce the standard deviation of each payoff observation, we achieve the same precision with fewer simulations.

4.3 Limitations of the Standard Estimator

While straightforward and unbiased, the standard Monte Carlo method has several important drawbacks that limit its practical usefulness:

- **Slow convergence:** The standard error decreases as $O(1/\sqrt{N})$. To halve the error, one must quadruple the number of simulations. Achieving “cent-level” accuracy (± 0.01) for a typical option may require $N > 10^6$ simulations.
- **High variance for out-of-the-money options:** When the strike K is large relative to the basket value, the vast majority of simulated paths produce a zero payoff. Only the rare paths where the basket finishes above the strike contribute to the price estimate. This creates a highly skewed payoff distribution with a few large positive values amid a sea of zeros, leading to high relative variance and unreliable estimates.

- **Wasted computation:** Many simulated paths produce a zero payoff, contributing no useful information about the option price but still consuming computational resources (random number generation, Cholesky decomposition, exponential function evaluation).
- **Noise in Greeks computation:** When computing sensitivities (delta, gamma) by finite differences of Monte Carlo prices, the simulation noise is amplified by the division by the small perturbation step h , leading to inaccurate Greeks unless an enormous number of paths is used.

These limitations motivate the variance reduction techniques developed in the following sections. The goal is to construct estimators that are still unbiased (they converge to the same price P) but have a much smaller variance, thereby achieving the same precision with far fewer simulations.

5 Reduced-Variance Estimator (Conditional Expectation)

5.1 Mathematical Foundation

The reduced-variance estimator is based on one of the most powerful and general variance reduction ideas: **conditional expectation**. The key mathematical insight is that replacing a random variable by its conditional expectation (given some information) preserves the overall expectation but reduces the variance.

Proposition 5.1 (Conditional Variance Reduction). *Let X be a random variable with finite variance, and let $\mathcal{G}$ be a sub- σ -algebra. Then:*

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X] \quad (\text{unbiasedness}) \quad (18)$$

$$\text{Var}(\mathbb{E}[X | \mathcal{G}]) \leq \text{Var}(X) \quad (\text{variance reduction}) \quad (19)$$

The proof follows immediately from the law of total variance:

$$\text{Var}(X) = \underbrace{\mathbb{E}[\text{Var}(X | \mathcal{G})]}_{\geq 0} + \text{Var}(\mathbb{E}[X | \mathcal{G}]) \quad (20)$$

Since the first term is non-negative, the variance of the conditional expectation is always less than or equal to the variance of the original variable. The reduction is greater the more informative $\mathcal{G}$ is — in other words, the more randomness we can “integrate out” analytically, the greater the variance reduction.

In our context, we take $X = \max(\alpha S_1(T) + \beta S_2(T) - K, 0)$ and $\mathcal{G} = \sigma(W_2(T))$, the sigma-algebra generated by the second Brownian motion. The conditional expectation $\mathbb{E}[X | W_2(T)]$ can be computed analytically, which is the key to the method.

5.2 Conditional Distribution of $S_1(T)$ Given $W_2(T)$

Given a simulated value of $W_2(T)$, we can compute $S_2(T)$ exactly, since $S_2(T) = S_2(0) \exp[(r - \sigma_2^2/2)T + \sigma_2 W_2(T)]$ is a deterministic function of $W_2(T)$. The remaining randomness lies entirely in $W_1(T)$.

We decompose $W_1(T)$ into its projection onto $W_2(T)$ and an independent residual:

$$W_1(T) = \rho W_2(T) + \sqrt{1 - \rho^2} \tilde{W}(T) \quad (21)$$

where $\tilde{W}(T)$ is independent of $W_2(T)$ and has the distribution $\mathcal{N}(0, T)$. This decomposition follows from the orthogonal projection of one Gaussian onto another.

Substituting this into the expression for $S_1(T)$, we obtain, conditionally on $W_2(T)$:

$$S_1(T) = \underbrace{S_1(0) \exp \left[\left(r - \frac{\sigma_1^2}{2} \right) T + \sigma_1 \rho W_2(T) \right]}_{\text{deterministic given } W_2(T)} \times \exp \left[\sigma_1 \sqrt{1 - \rho^2} \tilde{W}(T) \right] \quad (22)$$

The first factor is deterministic given $W_2(T)$; we denote it F (the conditional forward price). The second factor is log-normal with volatility $\sigma_1 \sqrt{1 - \rho^2}$. Conditionally on $W_2(T)$, $S_1(T)$ is therefore log-normally distributed with known parameters.

5.3 Computing the Conditional Expectation

The algorithm proceeds as follows for each simulation:

1. Simulate $W_2(T) = \sqrt{T} Y$ where $Y \sim \mathcal{N}(0, 1)$, and compute $S_2(T)$.
2. Define the conditional strike: $K' = \frac{K - \beta S_2(T)}{\alpha}$.
This is the effective strike for the call on $S_1(T)$ alone, once $S_2(T)$ is known.
3. Compute the conditional forward price of S_1 given $W_2(T)$:

$$F = S_1(0) \exp \left[\left(r - \frac{(\rho \sigma_1)^2}{2} \right) T + \sigma_1 \rho W_2(T) \right] \quad (23)$$

Note the adjusted drift: the term $\rho \sigma_1 W_2(T)$ is now part of the drift (since $W_2(T)$ is known), and the residual volatility is $\sigma_1 \sqrt{1 - \rho^2}$.

4. **Case 1:** $K' > 0$. The basket is not automatically exercised. We use a Black-Scholes-type formula with the conditional forward price F and the conditional volatility $\sigma_{\text{cond}} = \sigma_1 \sqrt{1 - \rho^2}$:

$$\text{price}_{\text{cond}} = F \Phi(d_1) - K' \Phi(d_2) \quad (24)$$

where

$$d_1 = \frac{\ln(F/K') + \sigma_{\text{cond}}^2 T/2}{\sigma_{\text{cond}} \sqrt{T}}, \quad d_2 = d_1 - \sigma_{\text{cond}} \sqrt{T} \quad (25)$$

This is simply the Black-Scholes formula applied to a call option with forward price F , strike K' , and volatility σ_{cond} .

5. **Case 2:** $K' \leq 0$. The basket is automatically in-the-money regardless of $S_1(T)$, because $\beta S_2(T) \geq K$. The conditional payoff is simply the intrinsic value:

$$\text{price}_{\text{cond}} = F - K' \quad (26)$$

This case occurs when $S_2(T)$ is sufficiently large relative to the strike, which is more likely for in-the-money options or high β values.

The final reduced-variance estimator is:

$$\hat{P}_{\text{red}} = \frac{1}{N} \sum_{i=1}^N e^{-rT} \alpha \text{price}_{\text{cond}}^{(i)} \quad (27)$$

5.4 Understanding the Variance Reduction Mechanism

The variance reduction stems from a fundamental smoothing effect. The original payoff $\max(\alpha S_1 + \beta S_2 - K, 0)$ is a non-differentiable, kinked function of the two Brownian motions. This non-smoothness is a source of high variance: small changes in the simulated path can cause the payoff to jump from zero to a large positive value (or vice versa).

By conditioning on $W_2(T)$, we replace this noisy, discontinuous payoff with its conditional expectation, which is a **smooth, deterministic function of $W_2(T)$** . The non-differentiable max function is replaced by a smooth analytical formula (the Black-Scholes-type expression), which naturally has lower variance. In essence, we have analytically integrated over all possible values of $\tilde{W}(T)$, removing the corresponding randomness from the estimator.

The effectiveness of the method depends critically on the correlation ρ :

- When ρ is close to ± 1 , most of the variability in $S_1(T)$ is explained by $W_2(T)$, so conditioning removes almost all the variance. The residual volatility $\sigma_1 \sqrt{1 - \rho^2}$ approaches zero, and the conditional expectation becomes nearly deterministic.
- When ρ is close to 0, the method still provides a reduction (by integrating over the residual randomness) but the improvement is less dramatic, since $W_2(T)$ carries little information about $S_1(T)$.

5.5 Convergence Analysis (Q6)

We compare the empirical variance and convergence of both estimators as a function of sample size, ranging from $N = 1,000$ to $N = 100,000$. This experiment allows us to verify two key theoretical predictions: (1) both estimators converge to the same price (unbiasedness), and (2) the reduced estimator has consistently lower variance.

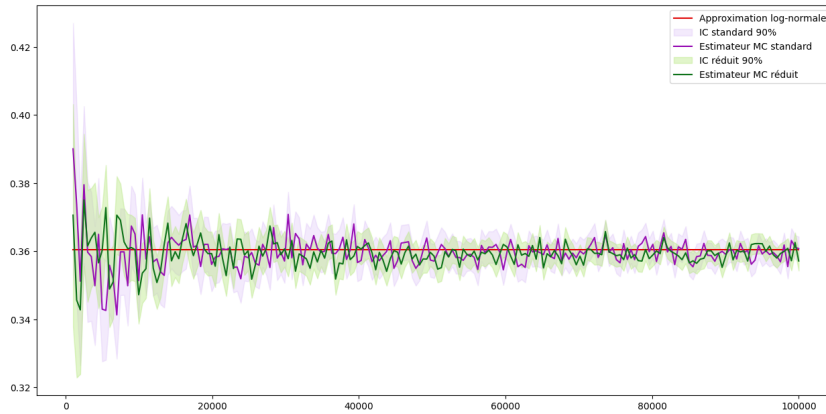


Figure 1: Empirical variance $\widehat{\text{STD}}^2/N$ of the standard (purple) and reduced (green) Monte Carlo estimators as a function of sample size N . The reduced estimator consistently achieves a variance several times smaller than the standard estimator, confirming the theoretical prediction of Proposition 1. The gap remains stable as N increases, indicating that the variance reduction factor is an intrinsic property of the method, not an artifact of sample size.

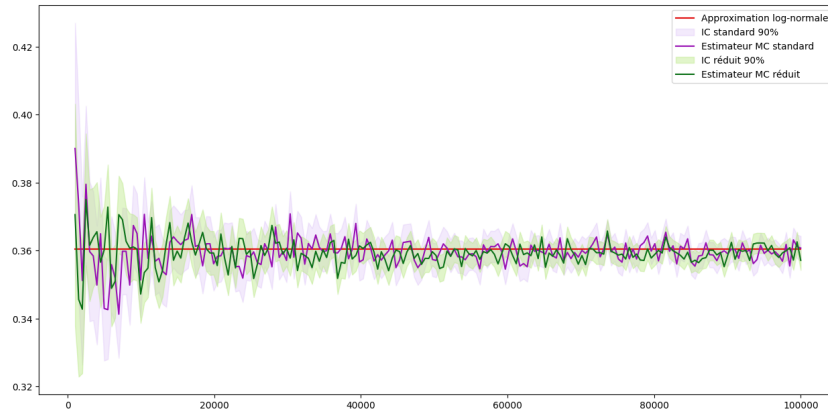


Figure 2: Convergence of the price estimators with 90% confidence intervals (shaded regions). The log-normal approximation (red line) provides a fixed reference. The reduced estimator (green) has visibly tighter confidence bands than the standard estimator (purple), meaning it converges to the true price faster. Both estimators fluctuate around the same value, confirming that the reduced estimator is unbiased.

The key observation from Figures 1 and 2 is that the reduced-variance estimator provides the **same expected price** (it is unbiased) but with a **much smaller variance**, which translates directly into tighter confidence intervals and faster convergence. In practical terms, to achieve a given precision, the reduced estimator requires far fewer simulations than the standard approach, leading to significant computational savings. This is the central result of the conditional expectation method: by doing a small amount of analytical work (evaluating the Black-Scholes formula) per simulation instead of simply computing the payoff, we extract substantially more information from each simulated path.

6 Impact of Correlation (Q7)

The correlation ρ between the two Brownian motions plays a crucial role in determining both the basket option price and the effectiveness of the variance reduction methods. In this section, we vary ρ from -0.99 to 0.99 and analyze its impact on the pricing and on the accuracy of the log-normal approximation.

6.1 Price Dependence on ρ

The basket option price depends on ρ through the effective basket volatility. When $\rho > 0$, the two assets tend to move in the same direction, increasing the dispersion of the basket value and thus the option price. Conversely, when $\rho < 0$, the assets partially cancel each other's movements, reducing the basket volatility and lowering the option price. This is a natural diversification effect: negatively correlated assets in the basket reduce the overall risk, which is precisely why basket options are popular as hedging instruments for diversified portfolios.

More precisely, the variance of the log-return of the basket depends on ρ through the cross-term:

$$\text{Var}(\alpha\sigma_1 W_1 + \beta\sigma_2 W_2) = \alpha^2\sigma_1^2 + \beta^2\sigma_2^2 + 2\alpha\beta\sigma_1\sigma_2\rho \quad (28)$$

This variance is minimized at $\rho = -1$ (perfect negative correlation) and maximized at $\rho = 1$ (perfect positive correlation). For our default parameters ($\alpha = \beta = 1$, $\sigma_1 = 0.35$, $\sigma_2 = 0.4$),

the cross-term coefficient is $2\alpha\beta\sigma_1\sigma_2 = 0.28$, which represents a significant portion of the total variance. This means that the option price is quite sensitive to changes in correlation.

At the extremes:

- $\rho \rightarrow -1$: The basket value $S_1(T) + S_2(T)$ becomes nearly deterministic (the two assets perfectly offset each other's random fluctuations), and the option price approaches zero because there is no volatility left to generate option value.
- $\rho \rightarrow +1$: The two assets move almost in lockstep, and the basket behaves like a single, highly volatile asset. The option price is then maximized, reflecting the high dispersion of possible outcomes.

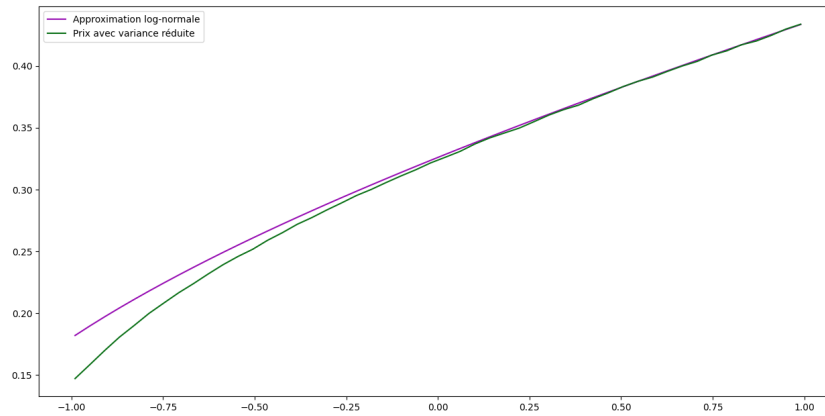


Figure 3: Basket option price as a function of correlation ρ . The price increases monotonically with ρ , reflecting the higher effective basket volatility. At $\rho = -0.99$, the price is near zero because the basket value is almost deterministic. At $\rho = 0.99$, the two assets move almost in lockstep, and the basket behaves like a single highly volatile asset.

6.2 Comparison of Estimators

Figure 4 compares the log-normal approximation with the reduced-variance Monte Carlo estimator across the full range of correlations.

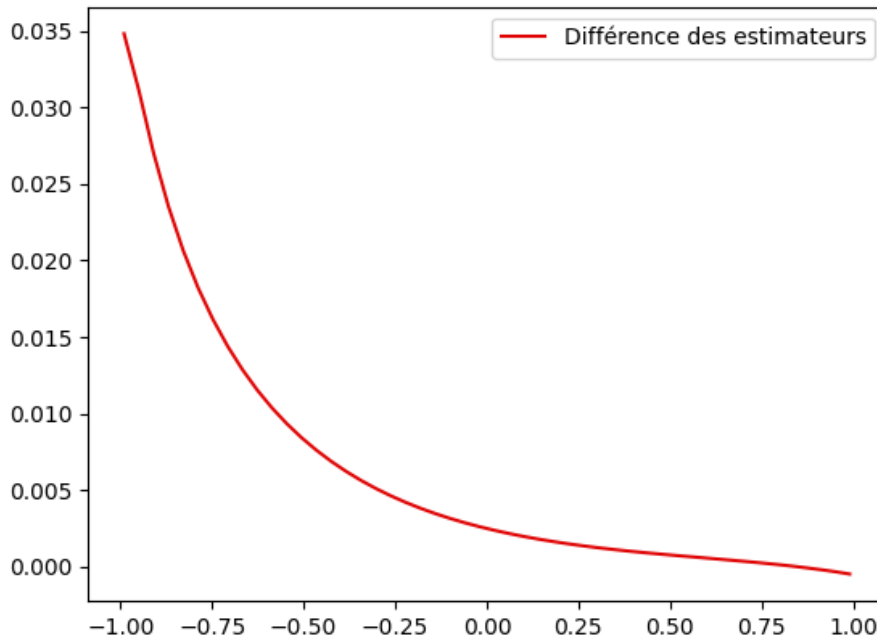


Figure 4: Difference $P_{BS} - P_{red}$ between the log-normal approximation and the reduced-variance Monte Carlo estimator as a function of ρ . The difference is small across most of the range but becomes noticeable at extreme correlations. This is because the log-normal approximation degrades when the basket distribution is far from log-normal — which happens when ρ is near ± 1 or when one weight dominates.

The near-zero difference in the central range of ρ (roughly $-0.5 < \rho < 0.8$) confirms that the log-normal approximation is a reasonable first-order estimate in this regime. However, the growing discrepancy at extreme values underscores the importance of Monte Carlo methods for accurate pricing when ρ is near ± 1 .

The error pattern can be understood as follows: at extreme correlations, the true basket distribution becomes increasingly non-log-normal (it develops a distinct shape that cannot be captured by matching only two moments). The log-normal fit, which matches the mean and variance but not the skewness or kurtosis, breaks down in these regimes.

7 Impact of Basket Weights (Q8)

We now investigate the effect of varying the basket weight α from 0.01 to 2, while keeping $\beta = 1$ fixed. This changes the relative contribution of each asset to the basket payoff and allows us to study how the pricing methods perform across different basket compositions.

7.1 Effect on Pricing

The basket weight α determines the relative importance of S_1 in the basket:

- When $\alpha \rightarrow 0$, the basket is dominated by S_2 and the option behaves essentially like a vanilla call on S_2 alone. The basket degenerates to a single-asset option, for which the Black-Scholes formula is exact.
- When $\alpha = 1$, the basket is equally weighted and both assets contribute equally.
- When $\alpha = 2$, the basket is weighted twice as much toward S_1 , and the option is more sensitive to the dynamics of S_1 .

The price of the basket option increases with α in our parameter configuration because adding more of S_1 (which has the same initial price as S_2 but slightly lower volatility) increases the expected basket value at maturity, making the option more likely to finish in the money. More precisely, the basket forward price is $\mathbb{E}[B(T)] = (\alpha S_1(0) + \beta S_2(0)) e^{rT}$, which increases linearly with α .

7.2 Estimator Comparison

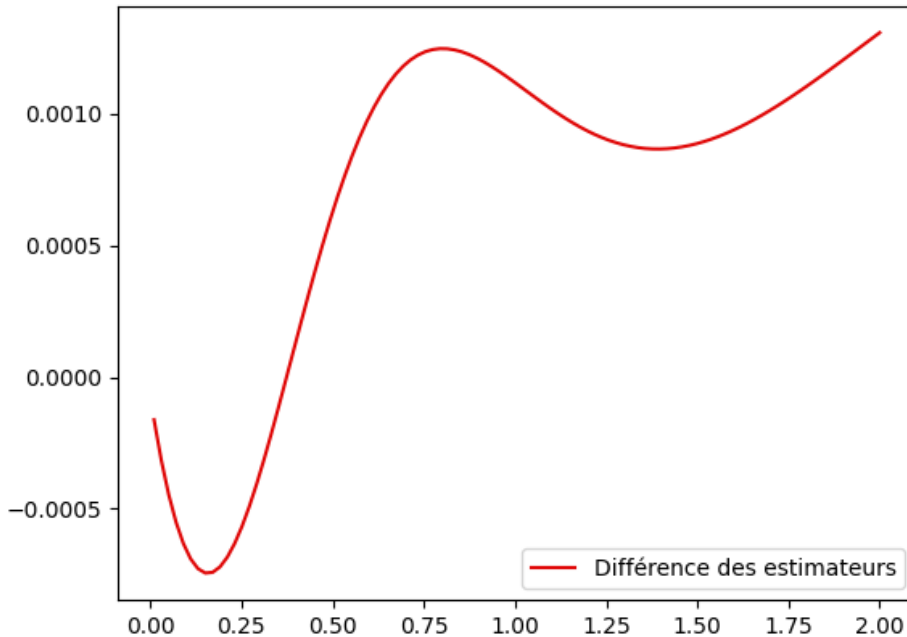


Figure 5: Difference between the log-normal approximation and the reduced-variance Monte Carlo estimator as a function of the basket weight α . The two estimators are closely aligned across the entire range, with a small but systematic difference. The approximation quality remains good because the log-normal fit captures the dominant behavior of the basket distribution for these parameter values.

The small and smooth nature of the difference curve in Figure 5 confirms that both methods produce consistent results across the full range of α . The residual error in the log-normal approximation is due to the skewness of the true basket distribution, which the log-normal model (characterized by only two parameters) cannot fully capture.

An important observation is that the approximation error does not vary dramatically with α for our parameter choices. This suggests that the weight structure of the basket has a relatively mild impact on the accuracy of the log-normal approximation, compared to the effects of correlation or strike price studied in the adjacent sections.

8 Impact of Strike Price (Q9)

The strike price K is perhaps the most important parameter determining the option's moneyness and, consequently, the effectiveness of different pricing methods. In this section, we vary K from 1 to 3 and analyze how the option price, the approximation error, and the variance reduction effectiveness all depend on the moneyness.

8.1 Price as a Function of K

The moneyness of the option has a dramatic effect on the price:

- For $K = 1$ (deep in-the-money), the option is almost certainly exercised at maturity, and its price is close to the intrinsic value $S_B(0) - Ke^{-rT} \approx 1.02$. The option behaves almost like a forward contract.
- For $K = 2$ (at-the-money), the option has the highest time value because the probability of finishing in or out of the money is roughly balanced. This is where the option is most sensitive to volatility and correlation.
- For $K = 3$ (deep out-of-the-money), the probability of exercise is very low (the basket must appreciate by 50% from its initial value) and the option price approaches zero. Only extreme scenarios produce a positive payoff.

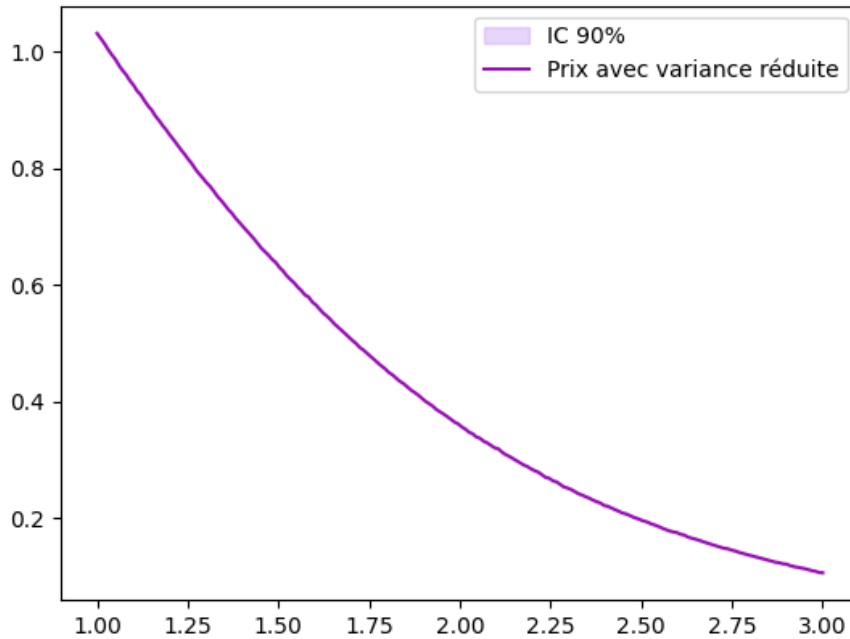


Figure 6: Basket option price as a function of strike K , comparing the log-normal approximation with the reduced-variance Monte Carlo estimator. Both curves overlap almost perfectly, confirming the accuracy of the reduced estimator. The price decreases monotonically with K , as expected for a call option.

8.2 Approximation Error

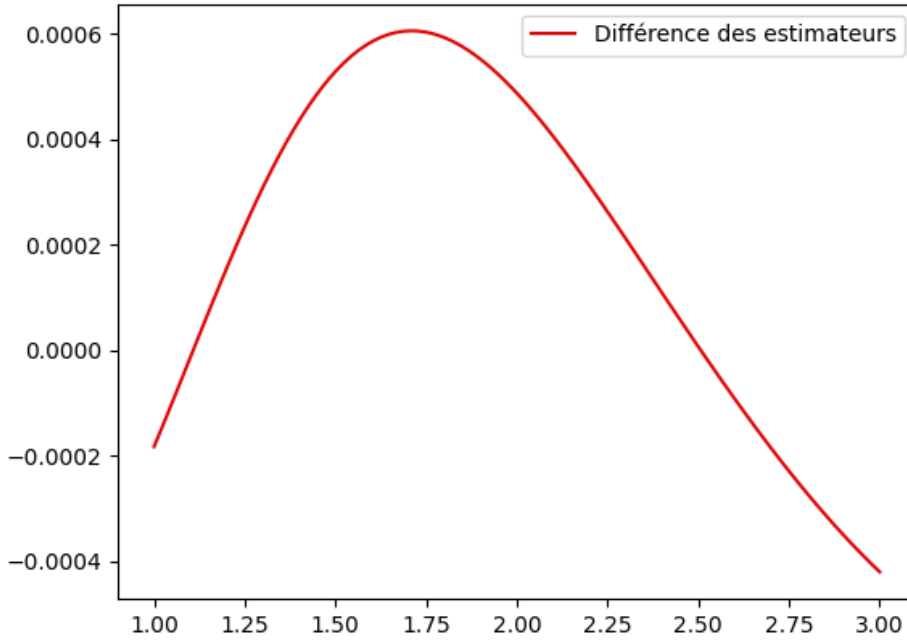


Figure 7: Difference $P_{BS} - P_{red}$ as a function of K . The log-normal approximation error grows at extreme strike values, particularly for out-of-the-money options ($K > 2$). This is because the tails of the true basket distribution differ from the log-normal assumption, and these differences are amplified in the option payoff when the strike is far from the money.

The error pattern in Figure 7 reveals an important structural feature of the log-normal approximation: it is most accurate near the money and degrades as the option moves deep in-the-money or deep out-of-the-money. This is a well-known phenomenon in quantitative finance, and it has a clear mathematical explanation.

The option payoff depends on the tail of the basket distribution — specifically, on the probability that the basket exceeds K at maturity. For deep out-of-the-money options ($K \gg S_B(0)$), this probability is determined by the far right tail of the distribution, where the true basket distribution (sum of log-normals) differs most from the fitted log-normal. The log-normal distribution has exponentially decaying tails, whereas the true distribution has heavier tails due to the dominance of the largest exponential term in the sum. This tail mismatch is what drives the growing approximation error.

This analysis highlights a general principle: **analytical approximations tend to be most accurate near the money and degrade for deep in-the-money or out-of-the-money options**. Monte Carlo methods, especially with variance reduction, provide a reliable and essentially bias-free alternative in all moneyness regimes.

9 Control Variate: Put-Call Parity (Q10)

9.1 Principle of Control Variates

The control variate method is one of the most versatile and widely used variance reduction techniques in Monte Carlo simulation. The idea is to exploit the knowledge of a correlated

quantity whose expected value is known analytically.

Suppose we want to estimate $\mathbb{E}[X]$ (the option price) via Monte Carlo, and we know a correlated random variable Y with known expectation $\mathbb{E}[Y]$. We construct the adjusted estimator:

$$\hat{X}_{CV} = X - c(Y - \mathbb{E}[Y]) \quad (29)$$

where c is a constant. This estimator is unbiased for any value of c , because $\mathbb{E}[Y - \mathbb{E}[Y]] = 0$. The variance of $\hat{X}_{CV}$ is:

$$\text{Var}(\hat{X}_{CV}) = \text{Var}(X) - 2c \text{Cov}(X, Y) + c^2 \text{Var}(Y) \quad (30)$$

This is a quadratic function of c that is minimized at:

$$c^* = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} \quad (31)$$

The optimal variance is then:

$$\text{Var}(\hat{X}_{CV}^*) = \text{Var}(X) (1 - \rho_{XY}^2) \quad (32)$$

where ρ_{XY} is the correlation between X and Y . This result shows that the variance reduction factor is $1/(1 - \rho_{XY}^2)$: the higher the correlation, the greater the reduction. For $\rho_{XY} = 0.95$, the variance is reduced by a factor of 10; for $\rho_{XY} = 0.99$, the factor exceeds 50.

9.2 Application: Put-Call Parity

A natural control variate for the basket call option is the corresponding **put option**, via the put-call parity relation. For European options, the put-call parity is an exact no-arbitrage identity:

$$C = P + \alpha S_1(0) + \beta S_2(0) - K e^{-rT} \quad (33)$$

where C is the call price and P is the put price. Instead of simulating the call payoff directly, we simulate the **put payoff**:

$$\text{payoff}_{\text{put}} = e^{-rT} \max(K - (\alpha S_1(T) + \beta S_2(T)), 0) \quad (34)$$

and deduce the call price via the parity relation:

$$\hat{P}_{\text{call}} = \overline{\text{payoff}_{\text{put}}} + \alpha S_1(0) + \beta S_2(0) - K e^{-rT} \quad (35)$$

The key observation is that the variance of the put payoff can be very different from the variance of the call payoff, even though the put-call parity guarantees that the prices are related by an exact identity. The reason is that the put and call have complementary payoff profiles:

- For in-the-money calls ($K < S_B(0) = 2$), the put is out-of-the-money. Most simulated paths produce a zero put payoff, with only a few paths contributing to the average. This sparse payoff structure leads to low variance.
- For out-of-the-money calls ($K > S_B(0)$), the situation is reversed: the call has few non-zero payoffs while the put has many, making the put payoff more variable.

9.3 When Is This Effective?

The put-control variate is most effective when the option is **in-the-money** ($K < S_B(0) = 2$ in our setup). In this case, the call has a large positive payoff with high probability, and the put has a small (often zero) payoff. The reduced number of non-zero put payoffs leads to a more stable estimator with lower variance.

Conversely, for out-of-the-money options, the put has many non-zero payoffs and the call has few, reversing the advantage. In this regime, the method may even increase variance relative to the standard Monte Carlo estimator.

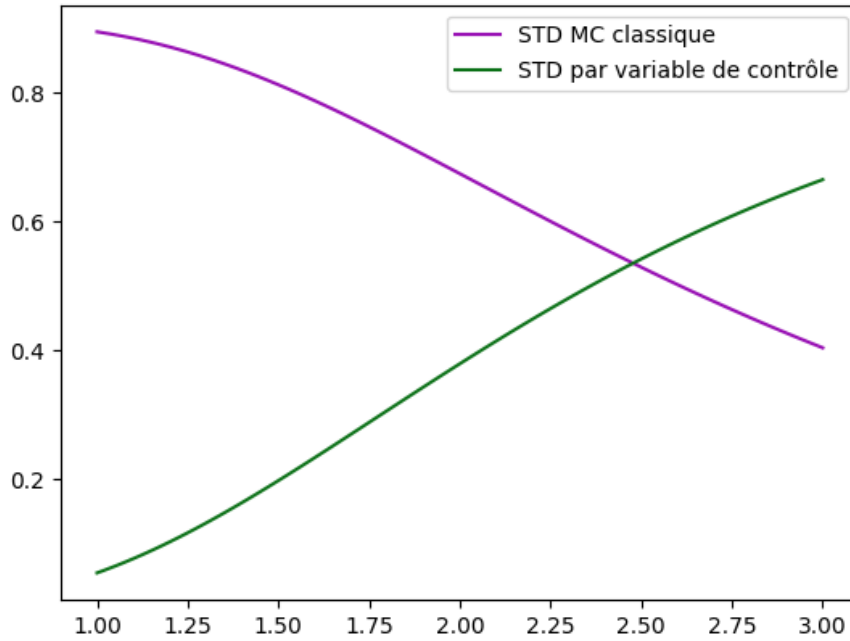


Figure 8: Standard deviation of the standard Monte Carlo estimator (purple) vs. the put-control variate estimator (green) as a function of strike K . The put control variate achieves lower variance for in-the-money options ($K < 2$), where the put payoff is sparse. For out-of-the-money options ($K > 2$), the standard method may be preferable. The crossover point occurs near the at-the-money strike $K = 2$.

Figure 8 clearly illustrates the moneyness-dependent effectiveness of the put-control variate. The two curves cross near $K = 2$ (the at-the-money strike), confirming the theoretical analysis. This result has a direct practical implication: the put-control variate should be used for pricing in-the-money basket calls, while an alternative method should be preferred for out-of-the-money options.

10 Delta Sensitivity (Q11)

10.1 Definition and Computation

The **delta** of an option measures the sensitivity of its price to changes in the underlying asset price. It is the most important Greek for hedging purposes: it tells a trader how many units of the underlying asset to hold in order to hedge the option's price risk.

For the basket option, the delta with respect to $S_1(0)$ is:

$$\Delta_1 = \frac{\partial P}{\partial S_1(0)} \quad (36)$$

Since the Monte Carlo price is not available in closed form as a function of $S_1(0)$, we use a **centered finite difference approximation**:

$$\Delta_1 \approx \frac{P(S_1(0) + h) - P(S_1(0) - h)}{2h} \quad (37)$$

where h is a small increment (we use $h = 0.01$). The centered (two-sided) difference is preferred over the one-sided difference because it has a truncation error of $O(h^2)$ instead of $O(h)$, providing better accuracy for the same step size.

A critical implementation detail is the use of **common random numbers**: we use the *same* set of random draws G for all three price evaluations ($P(S_1(0) + h)$, $P(S_1(0))$, and $P(S_1(0) - h)$). This ensures that the numerator of the finite difference measures the *true sensitivity* of the price to the perturbation, rather than being contaminated by random simulation noise. Without common random numbers, the finite difference would be dominated by the simulation noise, and the delta estimate would be essentially useless.

10.2 Results with Different Correlations

We compute the delta as a function of $S_1(0)$ for two correlation values: $\rho = 0.5$ and $\rho = -0.5$. The log-normal approximation provides an analytical reference for comparison.

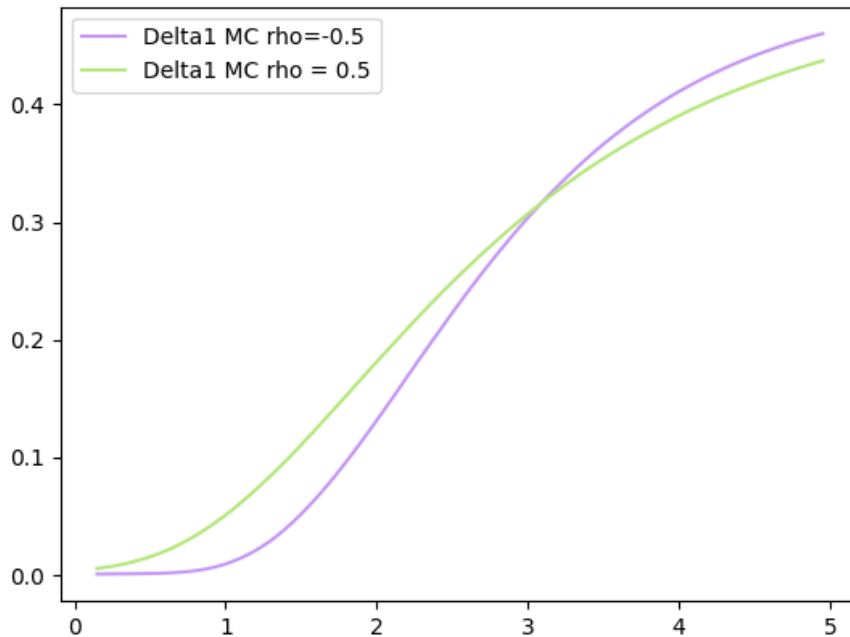


Figure 9: Delta Δ_1 as a function of $S_1(0)$ for $\rho = 0.5$ and $\rho = -0.5$, compared with the log-normal approximation. The delta increases from near 0 (deep out-of-the-money) to near α (deep in-the-money), following the classic S-shaped curve of a call option delta. The correlation affects the steepness of the transition: higher ρ leads to a steeper delta curve, reflecting the higher effective volatility.

The shape of the delta curve in Figure 9 is characteristic of a call option:

- **Deep out-of-the-money** ($S_1(0)$ small): The option is unlikely to be exercised, so $\Delta_1 \approx 0$. The option price is insensitive to small changes in $S_1(0)$.
- **At-the-money** ($S_1(0)$ near the threshold): The delta transitions rapidly from 0 to α , with the steepest slope corresponding to the region of maximum uncertainty about exercise. This is where the option is most sensitive to changes in the underlying price.
- **Deep in-the-money** ($S_1(0)$ large): The option is almost certainly exercised, and $\Delta_1 \approx \alpha$. The option behaves like a position of α shares in S_1 .

The correlation affects the steepness of the transition: higher ρ leads to a steeper delta curve because the effective basket volatility is higher, making the at-the-money region narrower. With lower correlation ($\rho = -0.5$), the transition is more gradual because the diversification effect reduces the effective volatility.

10.3 Approximation Error

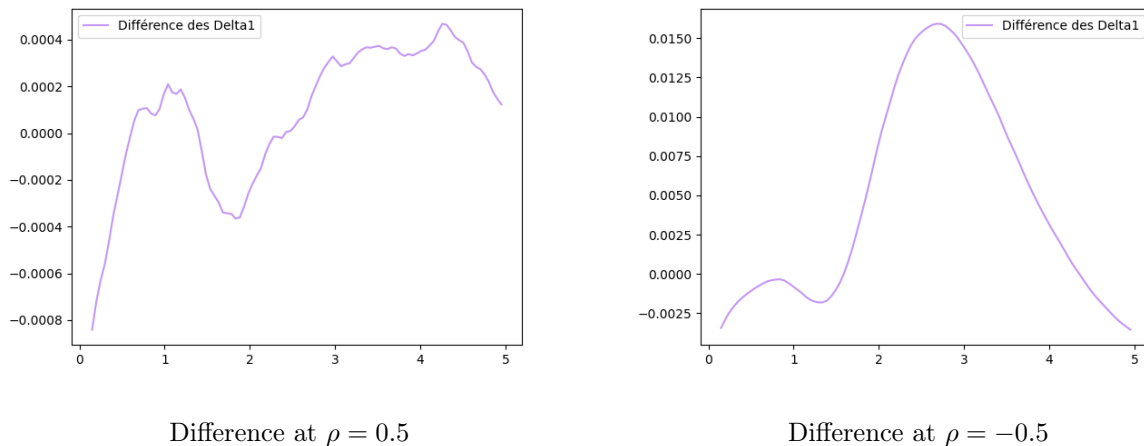


Figure 10: Difference between the analytical and Monte Carlo deltas for $\rho = 0.5$ (left) and $\rho = -0.5$ (right). The Monte Carlo estimates exhibit some noise due to numerical differentiation, which amplifies the simulation noise. Despite this, the overall shape and magnitude of the error are consistent, with the largest discrepancies occurring around the at-the-money region where the delta changes most rapidly.

Remark 10.1 (Numerical noise in finite-difference Greeks). *Computing derivatives numerically from Monte Carlo estimates is inherently noisy because the finite difference amplifies the random fluctuations in the price estimates by a factor of $1/(2h)$. With $h = 0.01$, this amplification factor is 50, meaning that the noise in the delta estimate is roughly 50 times larger than the noise in the price estimate. Using common random numbers (the same G for all evaluations) mitigates this effect by ensuring that the price difference captures the true sensitivity rather than random noise, but it cannot eliminate it entirely. For production applications, one would typically employ the **pathwise derivative method** (differentiating the payoff inside the simulation) or the **likelihood ratio method** (differentiating the density) to compute Greeks directly, without finite differences.*

11 Control Variate: Geometric Basket (Q12)

11.1 Motivation

The geometric basket provides a particularly powerful control variate for the arithmetic basket option, for two fundamental reasons:

1. **High correlation:** The geometric basket involves the same underlying assets as the arithmetic basket, and is therefore highly correlated with it. The geometric average $S_1^\alpha S_2^\beta$ and the arithmetic average $\alpha S_1 + \beta S_2$ are driven by the same random sources, so their payoffs tend to move together.
2. **Closed-form price:** Unlike the arithmetic basket, the geometric basket is a product of log-normal random variables, which is itself log-normally distributed. This means the geometric basket call option can be priced exactly using a Black-Scholes-type formula, providing the known expectation $\mathbb{E}[Y]$ needed for the control variate adjustment.

This combination of high correlation and known expectation makes the geometric basket an ideal control variate. The variance reduction factor $1/(1 - \rho_{XY}^2)$ can be very large when the correlation is close to 1, which is typically the case for the arithmetic and geometric basket payoffs.

11.2 Geometric Basket Definition

The geometric basket is defined as:

$$S_{\text{geo}}(T) = \exp(\alpha \ln S_1(T) + \beta \ln S_2(T)) = S_1(T)^\alpha S_2(T)^\beta \quad (38)$$

This is a product of log-normal random variables. Since the product of log-normals is log-normal (the log of the product is a sum of normals, which is normal), $S_{\text{geo}}(T)$ is itself log-normally distributed with parameters that can be computed analytically:

$$\mu = \alpha \left(\ln S_1(0) + \left(r - \frac{\sigma_1^2}{2}\right)T \right) + \beta \left(\ln S_2(0) + \left(r - \frac{\sigma_2^2}{2}\right)T \right) \quad (39)$$

$$\sigma = \sqrt{(\alpha^2 \sigma_1^2 + \beta^2 \sigma_2^2 + 2\alpha\beta\sigma_1\sigma_2\rho) T} \quad (40)$$

The parameter μ is the drift of the log-geometric basket (a weighted sum of the individual drifts), and σ is the effective volatility (which depends on the individual volatilities, the weights, and the correlation through a formula analogous to the portfolio variance formula).

11.3 Closed-Form Price

The geometric basket call option can be priced exactly using a Black-Scholes-type formula:

$$P_{\text{geo}} = e^{-rT} \left(e^{\mu + \sigma^2/2} \Phi(d_1) - K \Phi(d_2) \right) \quad (41)$$

with

$$d_1 = \frac{\mu - \ln K + \sigma^2}{\sigma}, \quad d_2 = \frac{\mu - \ln K}{\sigma} \quad (42)$$

This formula is exact (not an approximation) because the geometric basket is genuinely log-normal. The quantity P_{geo} is the known expectation needed for the control variate adjustment.

11.4 Regression-Based Coefficient Estimation

We simulate both the arithmetic and geometric basket payoffs using the same random numbers. The optimal control variate coefficient is estimated from the simulation itself via a simple regression:

$$\hat{c} = \frac{\widehat{\text{Cov}}(\text{payoff}_{\text{arithm}}, \text{payoff}_{\text{geo}})}{\widehat{\text{Var}}(\text{payoff}_{\text{geo}})} \quad (43)$$

This is equivalent to performing an ordinary least squares (OLS) regression of the arithmetic payoff on the geometric payoff. The coefficient $\hat{c}$ captures the strength and direction of the linear relationship between the two payoffs.

The adjusted estimator is then:

$$\hat{P}_{CV} = \frac{1}{N} \sum_{i=1}^N \left[\text{payoff}_{\text{arithm}}^{(i)} - \hat{c} \left(\text{payoff}_{\text{geo}}^{(i)} - P_{\text{geo}} \right) \right] \quad (44)$$

Note that using $\hat{c}$ instead of the true c^* introduces a small bias in the estimator (because $\hat{c}$ is itself a random variable correlated with the payoff). However, this bias is of order $O(1/N)$ and is negligible for the sample sizes we use ($N \geq 10^5$).

11.5 Results

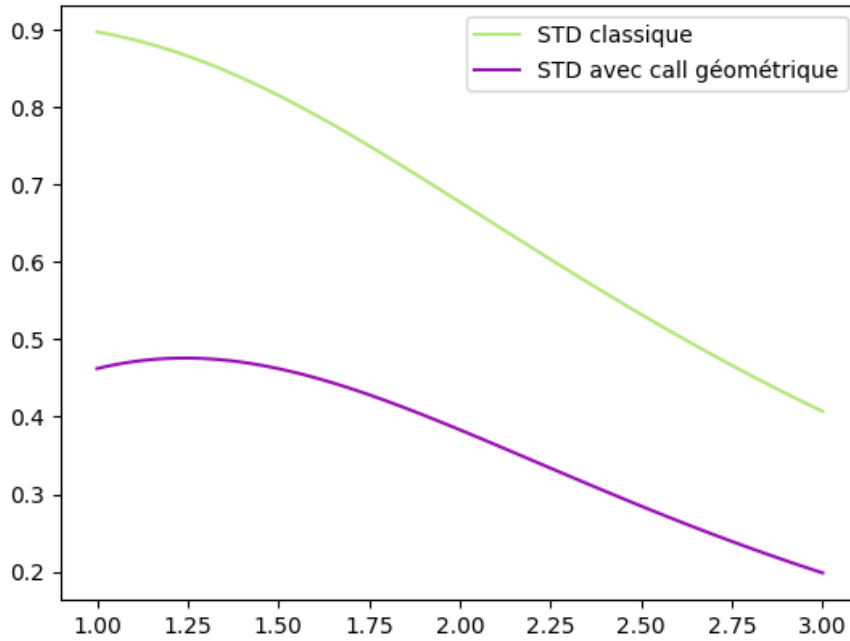


Figure 11: Standard deviation comparison: standard Monte Carlo vs. geometric basket control variate as a function of strike K . The geometric control variate achieves a dramatic variance reduction across all strikes, but the effect is most pronounced for out-of-the-money options where the standard estimator has the highest variance. The variance reduction factor can exceed 10x for deep out-of-the-money strikes, making this method essential for practical pricing in that regime.

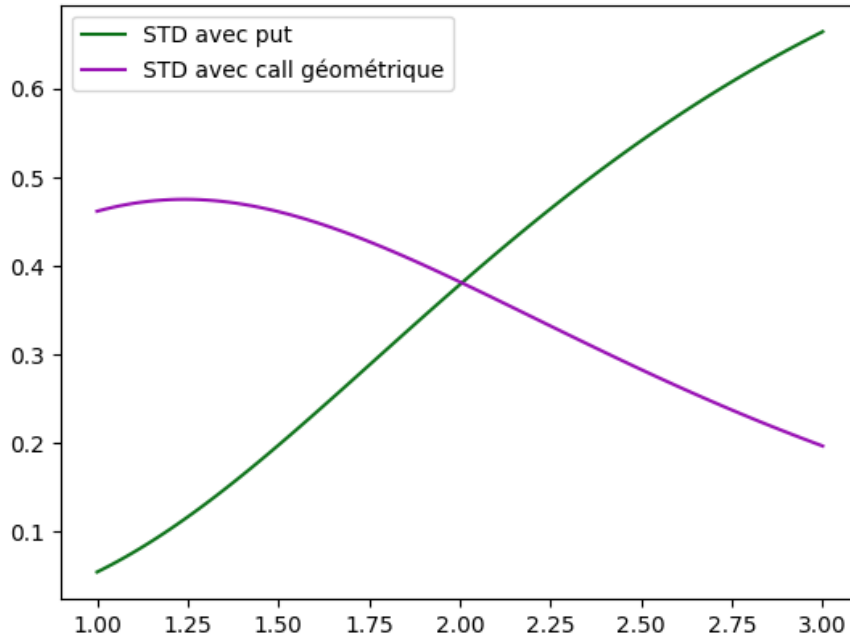


Figure 12: Comparison of the two control variate methods. The put-control variate (green) and geometric basket control variate (purple) are shown alongside the standard estimator. For in-the-money options ($K < 2$), the put-control variate achieves lower variance. For out-of-the-money options ($K > 2$), the geometric basket control variate dominates. This complementarity suggests that an optimal strategy would combine both methods, choosing the one best suited to the option's moneyness.

11.6 Interpretation and Comparative Analysis

The results in Figures 11–12 reveal a clear and important pattern: **no single variance reduction method dominates across all parameter regimes**. Instead, the methods are complementary, with each excelling in a different region of the parameter space:

- **In-the-money options** ($K < 2$): The put-control variate is most effective because the put payoff has few non-zero values, leading to low variance. The geometric basket control variate also performs well but is slightly less effective in this regime.
- **Out-of-the-money options** ($K > 2$): The geometric basket control variate is most effective. The arithmetic and geometric basket payoffs remain highly correlated even when most payoffs are near zero, because the two quantities are driven by the same underlying randomness. The standard estimator suffers most in this regime (most payoffs are zero), making the variance reduction especially valuable.
- **At-the-money options** ($K \approx 2$): Both control variate methods provide significant improvements over the standard estimator, with the geometric basket having a slight edge.

The **conditional expectation** method (Section 5) provides consistent variance reduction across all moneyness levels, making it the most robust choice when the moneyness regime is unknown or when one needs a single method that works well in all situations. However, for a specific known moneyness, the appropriate control variate can achieve even greater variance reduction.

12 Conclusion

This project provides a comprehensive study of Monte Carlo methods for pricing European basket options on two correlated assets. We implemented and compared five distinct pricing approaches — a log-normal approximation, a standard Monte Carlo estimator, a conditional-expectation-based reduced-variance estimator, and two control variate methods — analyzing each in terms of accuracy, variance, convergence, and robustness across different parameter configurations.

12.1 Summary of Methods

1. **Log-normal approximation (Black-Scholes):** Provides a fast analytical estimate by fitting a log-normal distribution to the basket through moment matching. It is useful as a benchmark and for quick sanity checks, but it is not exact and its error grows at extreme parameter values (high/low correlation, deep in/out-of-the-money). The approximation is most accurate near the money and for moderate correlations.
2. **Standard Monte Carlo:** The baseline numerical method. It is unbiased and converges at rate $O(1/\sqrt{N})$, but requires a large number of simulations to achieve high precision, especially for out-of-the-money options where the payoff distribution is highly skewed. It serves as the reference against which all variance reduction methods are measured.
3. **Conditional expectation (reduced variance):** Achieves significant variance reduction by analytically computing the conditional price given one of the Brownian motions. It is unbiased, robust across all parameter configurations, and requires no additional simulation cost. The variance reduction factor depends on the correlation ρ and can reach several orders of magnitude for extreme correlations. This is the most versatile method in our toolkit.
4. **Put-control variate:** Exploits the put-call parity relation to price the option via the put payoff, which has lower variance for in-the-money options. Simple to implement and effective in its domain, but only useful for a specific moneyness regime.
5. **Geometric basket control variate:** Uses the geometric basket — which admits a closed-form price — as a highly correlated control. Achieves the largest variance reduction for out-of-the-money options, where the arithmetic and geometric baskets remain strongly correlated. This method is complementary to the put-control variate.

12.2 Key Takeaways

Our analysis leads to several important conclusions for the practical pricing of basket options:

- Variance reduction techniques can reduce the standard deviation of Monte Carlo estimators by factors of 2–10x or more, translating directly into proportional computational savings. In some cases (conditional expectation with high correlation, geometric basket control variate for out-of-the-money options), the reduction exceeds two orders of magnitude.
- The choice of method should be guided by the **moneyness** of the option: put-control for in-the-money, geometric basket for out-of-the-money, and conditional expectation as a general-purpose approach when the moneyness is unknown or variable.
- The **log-normal approximation** is a useful first approximation for quick estimates and sanity checks, but should not be relied upon for precise pricing, especially at extreme parameter values where the approximation error can become significant.

- Computing **Greeks** (delta) via finite differences from Monte Carlo prices introduces numerical noise that is amplified by $1/(2h)$. Common random numbers mitigate this effect, but specialized methods (pathwise derivative, likelihood ratio) would be more appropriate for production applications requiring accurate sensitivities.
- The **correlation** ρ is a first-order determinant of both the option price and the effectiveness of the conditional expectation method. Accurate estimation of ρ is therefore crucial for both pricing and hedging.

12.3 Possible Extensions

Several directions could be explored to extend this work and increase its practical applicability:

- **Multi-asset baskets:** Extending to baskets with more than two assets, where the conditional expectation method generalizes by conditioning on all but one Brownian motion. The computational savings become even more significant in higher dimensions.
- **Quasi-Monte Carlo:** Replacing pseudo-random number generators with low-discrepancy sequences (Sobol, Halton, Niederreiter) for faster convergence rates of $O((\ln N)^d/N)$ in dimension d , combined with the variance reduction techniques studied here.
- **Importance sampling:** For deep out-of-the-money options, changing the probability measure to concentrate simulations in the region where the payoff is positive, then correcting with a likelihood ratio. This could further reduce variance beyond what the geometric basket control variate achieves.
- **Path-dependent basket options:** Extending the framework to price Asian basket options (payoff depends on the average basket value over time) or barrier basket options (payoff depends on whether the basket crosses a threshold), using the same variance reduction techniques.
- **Analytical Greeks:** Implementing the pathwise derivative method or the likelihood ratio method for exact Greeks computation without finite differences, eliminating the noise amplification problem identified in Section 10.
- **Stochastic volatility:** Extending the model to allow for stochastic volatility (e.g., Heston model) and studying how the variance reduction techniques perform in this more realistic setting.

Technical Appendix: Implementation

The complete implementation is available in `basket_options.py`, a single Python script containing all pricing functions and plotting code. The implementation uses the following libraries:

- **NumPy** — numerical computation, linear algebra (Cholesky decomposition), and array operations.
- **NumPy Random** — uniform random number generation, transformed to standard normals via the Box-Muller transform.
- **Matplotlib** — all visualizations and plots.
- **SciPy** (`scipy.stats.norm`) — accurate normal CDF evaluation used in the pricing functions.

Code structure. The script defines the following functions:

- `repartition_normale(x)` — Normal CDF approximation (Abramowitz & Stegun)
- `price_call_BS(...)` — Log-normal approximation (Black-Scholes)
- `call_MC(...)` — Standard Monte Carlo estimator
- `call_MC_red(...)` — Reduced-variance estimator (conditional expectation)
- `call_MC_red_control(...)` — Put-control variate estimator
- `call_MC_red_control_geo(...)` — Geometric basket control variate estimator
- `delta1_calc_MC(...)` — Delta via Monte Carlo finite differences
- `delta1_calc_approx(...)` — Delta via log-normal approximation
- `loi_normale(...)` — Box-Muller normal generation

All functions follow a consistent interface, accepting the model parameters and a pre-generated vector of standard normals G , which allows fair comparison between methods using identical random draws. This design choice is essential for the validity of our comparisons: by using the same random numbers across all methods, we eliminate the confounding effect of different random seeds and ensure that observed differences are due to the methods themselves, not to random variation.